

Modal Logic: Tableaux Examples

2026-02-18

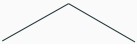
Example 2:

$$(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B) \text{ in } \mathbf{K}$$

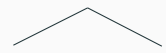
Example 2: $(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ in $\mathbf{K}$

Let's show this is a theorem in $\mathbf{K}$.

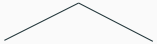
1. $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓ Assumption
2. $\mathbb{T} 1, \Diamond A \vee \Diamond B \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond(A \vee B) \rightarrow \mathbb{F} 1$

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|----|---|----------------------------|
| 1. | $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A \vee \Diamond B$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond(A \vee B)$ | $\rightarrow \mathbb{F} 1$ |
| |  | |
| 4. | $\mathbb{T} 1, \Diamond A \quad \mathbb{T} 1, \Diamond B$ | $\vee \mathbb{T} 2$ |

The disjunction branches. We can't do anything with the other line yet because there isn't yet a 1.1.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
		
4.	$\mathbb{T} 1, \Diamond A$ ✓ $\mathbb{T} 1, \Diamond B$	$\vee \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$

On the left branch, the True $\Diamond$ rule introduces world 1.1 with A true there.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
		
4.	$\mathbb{T} 1, \Diamond A$ ✓ $\mathbb{T} 1, \Diamond B$	$\vee \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
6.	$\mathbb{F} 1.1, A \vee B$	$\Diamond \mathbb{F} 3$

Now world 1.1 exists, so the False $\Diamond$ rule (line 3) applies: $A \vee B$ must be false at 1.1.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
$\swarrow \quad \searrow$		
4.	$\top 1, \Diamond A$ ✓ $\top 1, \Diamond B$	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
	⊗	

The False $\vee$ rule gives $\mathbb{F} 1.1:A$, which clashes with line 5. The left branch closes. Now work on the right branch.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$ ✓ $\mathbb{T} 1, \Diamond B$ ✓	$\vee \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
6.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
8.	⊗ $\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 4$

On the right branch, the True $\Diamond$ rule introduces world 1.2 with B true there.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
4.	$\top 1, \Diamond A$ ✓ $\top 1, \Diamond B$ ✓	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
8.	⊗ $\top 1.2, B$	$\Diamond \top 4$
9.	$\mathbb{F} 1.2, A \vee B$	$\Diamond \mathbb{F} 3$

World 1.2 now exists, so the False $\Diamond$ rule (line 3) applies again: $A \vee B$ must be false at 1.2.

1.	$\models 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \models 1$
3.	$\models 1, \Diamond(A \vee B)$	$\rightarrow \models 1$
4.	$\top 1, \Diamond A$ ✓ $\top 1, \Diamond B$ ✓	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\models 1.1, A \vee B$ ✓	$\Diamond \models 3$
7.	$\models 1.1, A$	$\vee \models 6$
8.	⊗ $\top 1.2, B$	$\Diamond \top 4$
9.	$\models 1.2, A \vee B$ ✓	$\Diamond \models 3$
10.	$\models 1.2, A$	$\vee \models 9$
11.	$\models 1.2, B$	$\vee \models 9$
	⊗	

Both branches close, so this **is** valid in K.